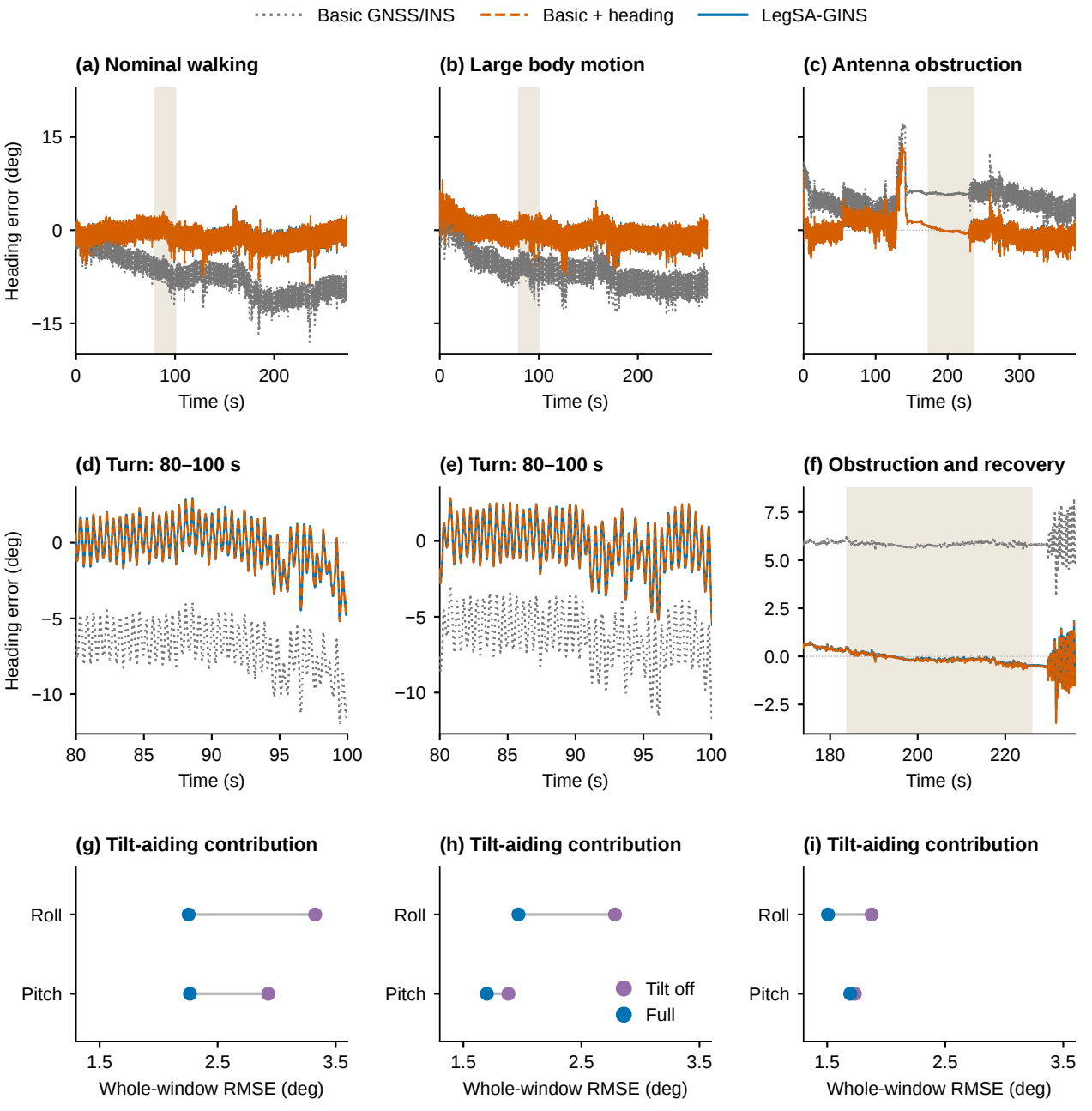

Basic GNSS/INS
Basic + heading
LegSA-GINS
(a) Nominal walking
(b) Large body motion
(c) Antenna obstruction
Heading error (deg)
15
0
−15
0
100
200
300
Time (s)
(d) Turn: 80–100 s
(e) Turn: 80–100 s
(f) Obstruction and recovery
−5
−10
80
85
90
95
100
7.5
5.0
2.5
0.0
−2.5
180
200
220
(g) Tilt-aiding contribution
(h) Tilt-aiding contribution
(i) Tilt-aiding contribution
Roll
Pitch
Tilt off
Full
1.5
2.5
3.5
Whole-window RMSE (deg)